

SEDAR 57 U.S. Caribbean Spiny Lobster

DATA WORKSHOP RESEARCH RECOMMENDATIONS

LIFE HISTORY RESEARCH RECOMMENDATIONS

Stock Definition and Description

- Research on stock structure is needed, particularly as it relates to connectivity caused by larval dispersal.
- Encountering the right habitat is important for survival of juvenile lobster recruits. Research should be conducted to explore effects of sargassum, water quality, coastal development, and mangrove root communities on the availability and quality of habitat for juvenile spiny lobsters.

Length-Weight Conversions

- Explore plausibility of cause and effect mechanisms that may lead to temporal growth variation.

Discard Mortality

- Investigate potentially unaccounted for discards in the self-reported commercial logbook data to be able to quantify the number of lobster discarded dead, as well as the number of lobster discarded alive.
- Research aimed at quantifying post-release mortality (including post-release predation) of spiny lobster to better understand and propose mechanisms that could potentially mitigate mortality among lobsters that are discarded.

FISHERY-INDEPENDENT DATA SOURCES

- Development of fishery-independent surveys that are specifically designed for spiny lobster, which would require considerable planning regarding data priorities (e.g., relative abundance versus length), the life stage to target (e.g., adult, juveniles, or larvae), type of gear, sampling design, temporal and spatial resolution, and the availability of funds. In addition to discussing field sampling, planning of how best to record and store data would be beneficial to future analyses and stock assessments.
- Research aimed at identifying correlations between larval and juvenile abundance from the SEAMAP-C surveys and lobster landings could assist in determining the relationship between juvenile abundance and adult abundance (e.g., Butler et al. 2010).

COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

- General data improvements are recommended, including continued reporting of specific gear categories (eg. different types of diving).
- Investigate the sensitivity of stock assessment results to landings data associated with high uncertainty.
- Investigate improvements or alternatives to past correction factors in Puerto Rico (2005 in particular).
- Continue SEFSC funded commercial landings validation studies in Puerto Rico and begin similar surveys in the US Virgin Islands.
- General data improvements are recommended, including encouraging complete reporting of discards.

RECREATIONAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

- Permanent programs that quantify the recreational effort and landings in the US Caribbean are needed. The results of recent pilot studies (Valle-esquivel and Trumble 2016 and Goedeke et al. 2016) should be used to develop future surveys.

STAKEHOLDER INPUT RESEARCH RECOMMENDATIONS

- Continue comprehensive bio-socio-economic database of events, compile references and time series of quantitative as available.
- Identify significant EBM quantitative socioeconomic indicators (ex. gravity of the market, network market analyses, population growth, tourism, poaching).
- A Caribbean-specific staff for data statistics and assessments was recommended to aid in establishing and maintaining high expertise of accessing and analyzing past, current, and future data collected in the region.

ASSESSMENT WORKSHOP RESEARCH RECOMMENDATIONS

- *Independently estimate availability/selectivity.* There are three main parameters currently estimated in the SEDAR 57 stock assessment for Caribbean spiny lobster in Puerto Rico. They are R_0 , selectivity, and initial F . Selectivity, as it is used in the model, is both a combination of contact selectivity e.g. selection created by contact with the gear itself such as trap opening diameter and availability which might be a function of depth and habitat. In the base model, selectivity is assumed to be dome-shaped based on information from STT. This is an important assumption since it affects the estimates of both initial F and R_0 . Knowing more about gear selectivity or having a survey that can provide the underlying size distribution of all lobster in the population, across all habitat and depths, and not just those targeted by the commercial fishery, could greatly improve the stock assessment in Puerto Rico and help determine the appropriateness of the assumed selectivity pattern.
- *Collect data on recreational landings.* The magnitude of recreational removals of spiny lobster on each island platform is unknown.

- *More basic biological studies* to improve understanding of key life history processes such as growth, length/age at maturity, fecundity, and their spatial variability.
- *Improve data on commercial landings and catch and effort.* Concerns of misreporting should be investigated and corrected where practicable. Commercial catch and effort may provide CPUE indices in the future.

REVIEW PANEL RESEARCH RECOMMENDATIONS

The recommendations provided by the Data and Assessment workshops are presented in Appendix 1 of this Consensus Summary Report. Priorities are provided against each of the recommendations although several of the recommendations could be captured in a single research program. The following are the key areas that the RP suggest be the focus of initial research:

- (i) Development of an abundance index. An index based on fishery dependent data is complicated by the variety of gears used (fish traps, lobster pots, and various forms of diving), but this would probably be the most cost effective and provide a historical index. This can be done through either standardizing fishery dependent or fishery independent methods.
- (ii) It was indicated that there are a number of Marine Protected Areas (MPAs) that have been established for a period of time and are policed for illegal activity. These MPAs represented regions that may provide the closest approximation of an unfished resource. Providing that permits can be obtained, these locations provide opportunities to obtain selectivities that can be used to validate, among other things, the dome-shaped selectivities used in the assessment and an abundance index for an unexploited stock. Size frequency distributions would also assist in determining appropriate growth parameters and should be combined with tagging to improve the estimates of growth.
- (iii) An uncertainty in the catch data relates to unreported catch in the form of recreational, illegal and unregulated catch. S57_RD_11 and S57_RD_23 both suggest that unlicensed and recreational fishing are issues for PR. Gaining an understanding of the magnitude of these catches and thus incorporating all forms of harvest in the total catch estimation is a high priority. Discards (identified in the logbook) and their fate should also be investigated and incorporated.
- (iv) The RP suggested that the uncertainty within the expansion factors should be explored in detail and be based on species specific expansion factors. Efforts should be targeted to improving reporting rates.
- (v) Re-estimate growth parameters using old and new data with more modern techniques.